

Figure 1. One platform, two cooperating modes, in a live energy OT environment

SOC Mode = deception-based active defence. Research Mode = offline analyst copilot. They share one data layer: live attacker behaviour informs research, research insight informs defence.

Energy operator OT environment (Purdue model)

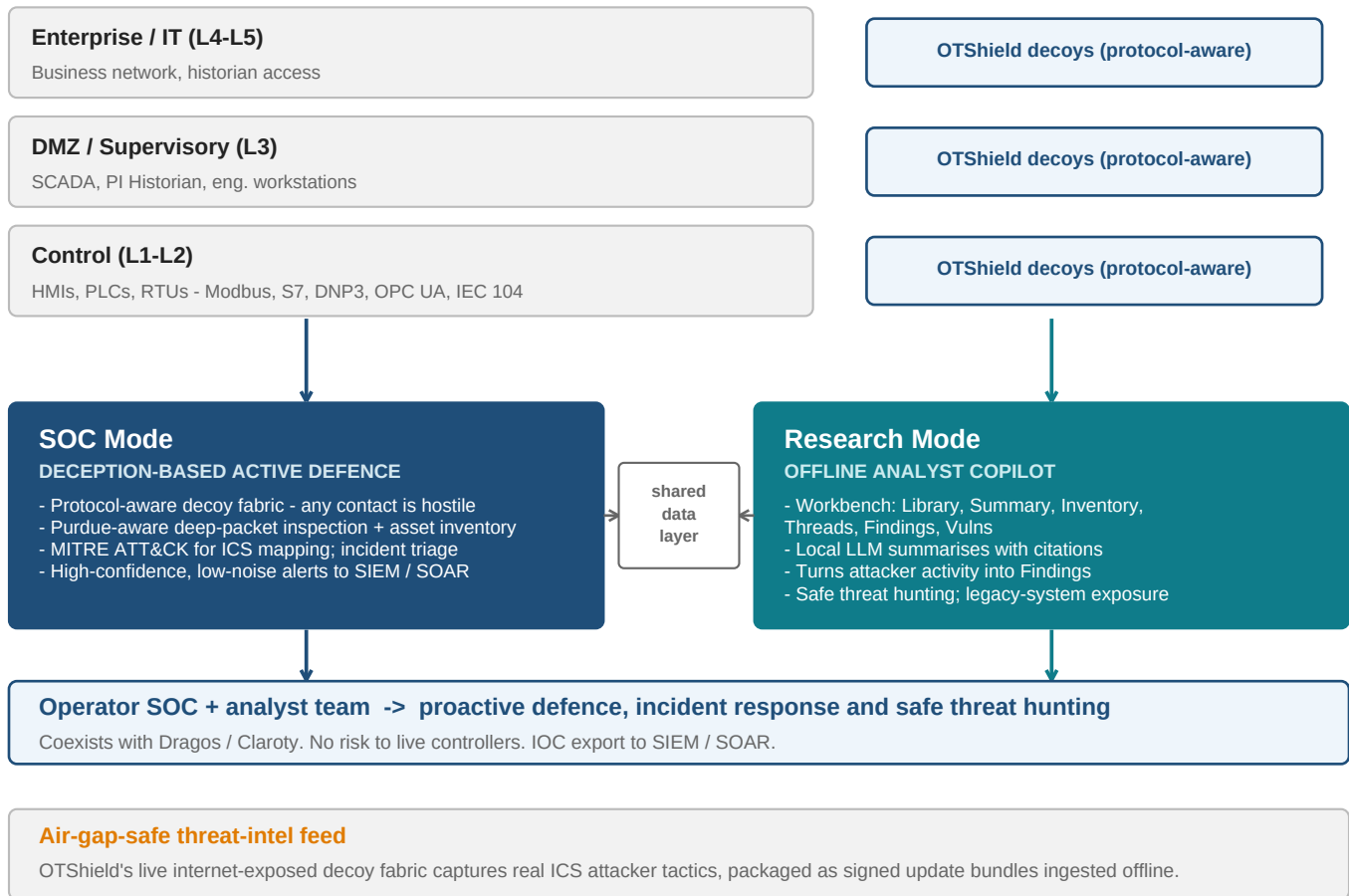
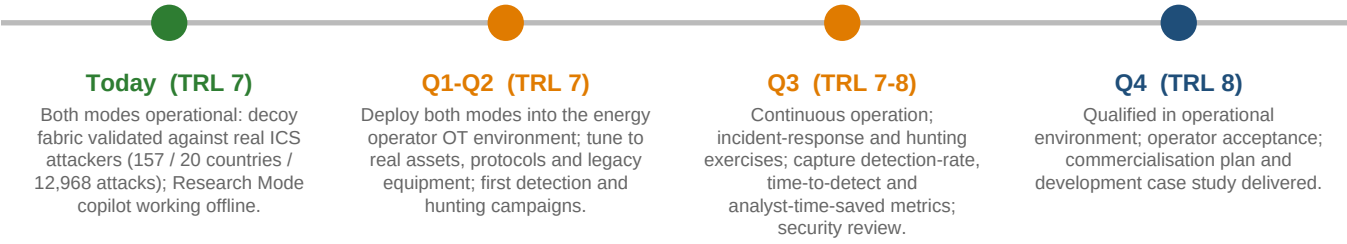


Table 1. Mapping to the four competition challenges

Challenge	Mode	How OTShield delivers it in this project
1. Proactive CNI defence & incident response	SOC Mode	High-confidence decoy detections + DPI; alerts to SIEM/SOAR; triage and response with the operator SOC team.
2. Detection & eviction of sophisticated actors	SOC Mode	Surfaces protocol-aware intent (unauthorised writes, interlock overrides, remote discovery) mapped across ATT&CK for ICS.
3. Threat hunting for CNI organisations	Research Mode	Copilot lets analysts query detections, manuals and asset data together and follow leads safely with zero risk to live controllers.
4. Legacy system vulnerabilities in CNI	Both modes	SOC Mode inventories and risk-scores legacy assets; Research Mode copilot reasons over them to show how they could be reached and abused.

Figure 2. Readiness: TRL 7 today -> TRL 8 at project end



Why it meets the challenge

One integrated platform - SOC Mode for deception-based active defence, Research Mode for the analyst copilot - is proven, accepted and qualified inside a real UK energy operator, delivering all four challenge areas and ready to scale across the sector.